

CEE 406 Pavement Design I
Department of Civil and Environmental Engineering
University of Illinois at Urbana-Champaign
Homework #1
Due: 9/12/2024

1. Use Microsoft PowerPoint to draw and label cross-sections of a typical flexible pavement and a typical rigid pavement. Describe the function of each of these layers. Please include typical thicknesses.
2. What is the “mechanistic-empirical pavement design method”? Which component is mechanistic, and which one is empirical? Is it possible to develop a totally mechanistic pavement design method? Why?
3. What is the difference between a seal coat, tack coat, and a prime coat? Which requires the use of less viscous asphalt?
4. Describe the mechanics of pumping. What are the main consequences and how could it be fixed?
5. What are the 2 kinds of rutting (relating to pavement layers)? Explain what happens in both. Explain the design methods that have been used to control rutting.
6. Describe the difference between a single, tandem, and tridem axle. For the same load on each axle, tire design, and tire pressure, which axle does the least damage to pavement? Explain your answer.
7. What is the impact of tire pressure on flexible pavements. How does tire pressure affect trucking operation.